

Trigonometric Equations — Problem Set

Note: This used to be part of Year 12 Extension 1 in the 2017 Syllabus.

As such, all examinations mentioned here are Year 12 Trial Papers.

t-formulae which used to be in Year 12 Extension 1 in the 2017 syllabus is moved to Year 12 Extension 2 in the new 2024 Syllabus

Warm-up

1. Solve $\sin x = \frac{1}{2}$ for $0 \leq x \leq 2\pi$.
2. Express $\sin x + \cos x$ in the form $R \sin(x + \alpha)$ where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.
3. State the maximum and minimum values of the function $f(x) = 5 \cos(x - \frac{\pi}{6})$.
4. If $\tan \alpha = \sqrt{3}$ and α is acute, find the value of α in radians.
5. Expand $R \cos(x + \alpha)$ using the compound angle formula.
6. Solve $2 \cos \theta + 1 = 0$ for $-\pi \leq \theta \leq \pi$.
7. Find the value of R (where $R > 0$) for the following expressions:
 - a) $3 \sin x + 4 \cos x$
 - b) $\sin x - \cos x$
 - c) $\sqrt{5} \cos x + 2 \sin x$
8. Determine the value of α in degrees ($0^\circ < \alpha < 90^\circ$) such that $\sin x + \sqrt{3} \cos x = R \sin(x + \alpha)$.
9. Solve $\sin 2x = \frac{1}{2}$ for $0 \leq x \leq \pi$.
10. Express $5 \cos \theta - 12 \sin \theta$ in the form $R \cos(\theta + \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.
11. Solve $\cos(x - \frac{\pi}{4}) = \frac{\sqrt{2}}{2}$ for $0 \leq x \leq 2\pi$.

Skill-building

1. Express $5 \sin x + 12 \cos x$ in the form $R \sin(x + \alpha)$ where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.
2. Solve $\sqrt{2} \sin x - \sqrt{2} \cos x = 1$ for $0 \leq x \leq 2\pi$ by first writing the LHS in the form $R \sin(x - \alpha)$
3. Find the solutions to the equation $\cos 2\theta = \frac{1}{2}$ where $0 \leq x \leq 2\pi$.
4. Express $2 \cos x - \sin x$ in the form $R \cos(x + \alpha)$ and hence state its maximum value.
5. Solve $2 \sin^2 \theta + \sin \theta - 1 = 0$ for $0 \leq \theta \leq 2\pi$.
6. Rewrite $4 \sin \theta - 3 \cos \theta$ in the form $R \sin(\theta - \alpha)$.
7. Solve $\sin 2x = \cos x$ for $0 \leq x \leq 2\pi$.

8. Given $f(x) = \sqrt{3}\cos x + \sin x$, find the first positive x -intercept.
9. Solve $3\tan^2 x = 1$ for $0 \leq x \leq 2\pi$.
10. Express $\cos x - \sin x$ in the form $R\cos(x + \alpha)$ and solve $\cos x - \sin x = \frac{1}{\sqrt{2}}$ for $0 \leq x \leq 2\pi$.

Easier Exam Questions

1. (Pymble 2025 Q12a)

Write $\sqrt{3}\cos x - 3\sin x$ in the format $R\cos(x + A)$ where $R > 0$ and $0 < A < \frac{\pi}{2}$. **2**

2. (Penrith 2023 Q13b)

Solve $3\sin x - \sqrt{3}\cos x = \sqrt{3}$ for $0 \leq x \leq 2\pi$ **3**

3. (St George Girls 2024 Q12a)

(i) Express $\sqrt{12}\sin x + 2\cos x$ in the form $R\sin(x - \alpha)$, where $R > 0$ and $0 \leq \alpha < 2\pi$. **3**

(ii) Hence, or otherwise, solve $\sqrt{12}\sin x + 2\cos x = 2$, for $0 \leq x \leq 2\pi$. **1**

4. (Frensham 2021 Q12a)i)

Write the expression $\sqrt{3}\sin x + \cos x$ in the form $R\sin(x + \alpha)$, **2**
where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$

5. (Hunters Hill 2022 Q12a)

Find all values of θ , such that $-\pi \leq \theta \leq \pi$, that satisfy the equation **3**

$$3\sin \theta = \sqrt{3}\sin 2\theta.$$

6. (Trinity Grammar 2024 Q11b)

(i) Rewrite $\cos x - \sqrt{3}\sin x$ in the form $R\cos(x + \alpha)$, where $0 \leq \alpha \leq \frac{\pi}{2}$, $R > 0$. **2**

(ii) Hence, or otherwise, solve $\cos x - \sqrt{3}\sin x = 1$ for $0 \leq x \leq 2\pi$. **2**

7. (Sydney Grammar 2021 Q12e)

(i) Express $3\cos x + 2\sin x$ in the form $R\cos(x - \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. **2**

(ii) Hence, or otherwise, solve $3\cos x + 2\sin x = \sqrt{13}$, for $0^\circ \leq x \leq 360^\circ$. Give your answer in degrees correct to two decimal places. **2**

8. (St George Girls 2022 Q12d)

By expressing $\sqrt{3}\sin \theta - \cos \theta$ in the form $R\sin(\theta - \alpha)$ where $R > 0$,
solve the equation $\sqrt{3}\sin \theta - \cos \theta - 1 = 0$ for $[0, 2\pi]$. **3**

9. (Caringbah 2024 Q11b)

(i) Express $\cos x - \sqrt{3}\sin x$ in the form $R\cos(x + \alpha)$ where **2** $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.

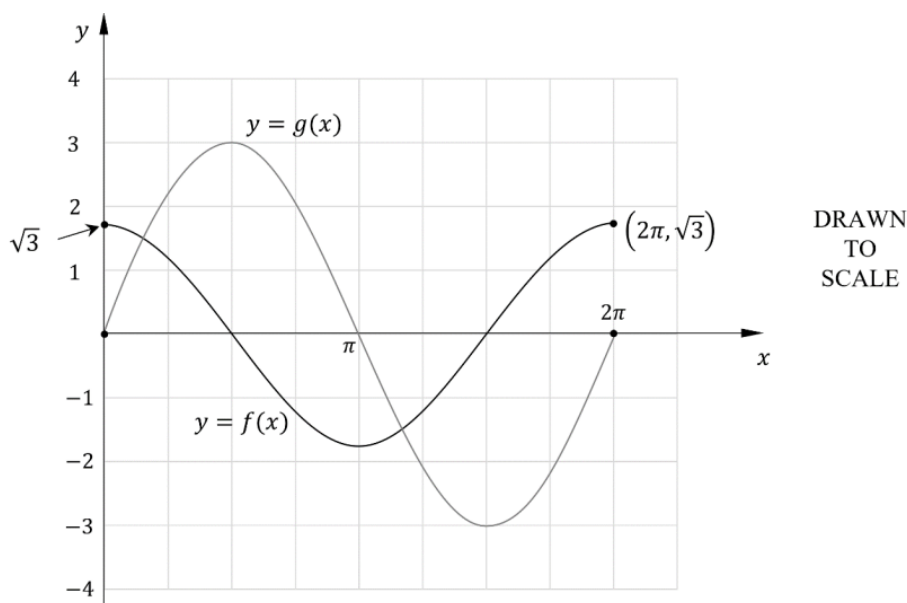
(ii) Hence, solve $\cos x - \sqrt{3}\sin x = -2$ for $0 \leq x \leq 2\pi$ **2**

10. (Riverview 2021 Q2a)
- (i) Write the expression $\sin x + \sqrt{3} \cos x$ in the form $R \sin(x + \alpha)$, where $R > 0$ and $0 \leq x \leq \frac{\pi}{2}$ 2
- (ii) Hence, solve the equation $\sin x + \sqrt{3} \cos x = 1$ in the domain $[0, 2\pi]$ 2
11. (Fort St 2024 Q11e) (i) Express $\cos x - \sqrt{3} \sin x$ in the form $R \cos(x + \alpha)$ for $R > 0$ and α acute. 2
- (ii) Hence, solve $\cos x - \sqrt{3} \sin x = 2$ for $0 \leq x \leq 2\pi$. 2
12. (Barker 2025 Q12b)
- (i) Express $\sqrt{3} \cos x - \sin x$ in the form $R \cos(x + \alpha)$ where $0 \leq \alpha \leq \frac{\pi}{2}$. 1
- (ii) Hence, or otherwise, solve $\sqrt{3} \cos x - \sin x = 1$ for $0 \leq x \leq 2\pi$. 2

Harder Exam Questions

1. (Caringbah 2023 Q13d)

The graphs of the trigonometric curves $y = f(x)$ and $y = g(x)$ are shown below. 3



Find the equation of the curve $y = f(x) + g(x)$, expressed in the form $y = R \cos(x - \alpha)$, where α is in radians.

2. (Merewether 2025 Q11f)

Solve $\cos 2\theta = \cos^2 \theta + \cos \theta + 1$ for $0 \leq \theta \leq 2\pi$ 2

3. (Penrith 2022 Q13c)

The equation $I(t) = 4 \sin 0.02t + 8 \cos 0.02t$ measures the current, I amperes, in a particular circuit after t seconds.

- (i) Express the function in the form $I(t) = A \sin(at + b)$, for $0 \leq b \leq \frac{\pi}{2}$. 3
- (ii) When does the current first reach its maximum value? 3

4. (Abbotsleigh 2025 Q11e)

Solve $\sqrt{2} \cos\left(x + \frac{\pi}{4}\right) + \sqrt{2} \cos\left(x - \frac{\pi}{4}\right) = 1$ for $-\pi \leq x \leq \pi$.

2

5. (Riverview 2025 Q11d)

The equation $I(t) = 4 \sin 0.02t + 8 \cos 0.02t$ measures the current, I amperes, in a particular circuit after t seconds.

(i) Express the function in the form $I(t) = A \sin(at + b)$, for $b = \left[0, \frac{\pi}{2}\right]$

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(ii) When does the current first reach its maximum value to the nearest second?

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6. (Sydney Boys 2020 Q13b)

Solve $\frac{\sin x + \cos x}{\sqrt{2}} = \frac{\sqrt{3}}{2}$ for $0^\circ \leq x \leq 360^\circ$.

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7. (Glenwood 2023 Q12b)

Express $9 \sin x + 40 \cos x$ in the form $A \sin(x + \alpha)$ where $0 \leq \alpha \leq \frac{\pi}{2}$, and

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hence or otherwise solve $9 \sin x + 40 \cos x = 6$ for $0 \leq x \leq \pi$.

(Leave your answer for x and α in radians, correct to three significant figures).

8. (Sydney Boys 2025 Q11c)

Solve $\sin \theta + \cos \theta = -1$ for $0 \leq \theta \leq 2\pi$.

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9. (Ascham 2023 Q13a)

i) Show that $\tan 2x + \tan x = \frac{\sin 3x}{\cos 2x \cos x}$.

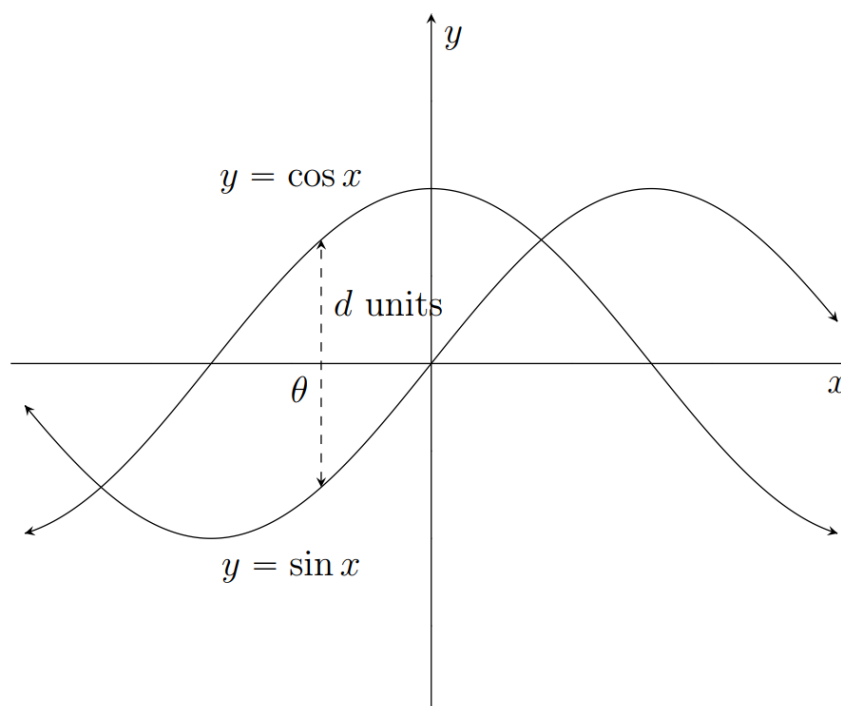
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ii) Hence solve the equation $\tan 2x + \tan x = 0$ for $0 \leq x \leq \frac{\pi}{2}$.

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10. (Sydney Girls 2025 Q13d)

The graphs of $y = \cos x$ and $y = \sin x$ are sketched below.



The dashed line indicates the greatest vertical distance, d units, between the two curves.

The x -coordinate of the dashed line is θ .

(i) Find d , giving reasons for your answer. **1**

(ii) Find θ , giving reasons for your answer. **1**